Econ 828: International Trade II
Spring 2026

Wednesdays
8.30-11.00hrs
Social Sciences 113
January 14th to April 15th: 13 classes

Instructors: Felix Tintelnot (first half) and Federico Huneeus (second half)
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Description

This is a second-year PhD course in advanced topics in international trade. It corresponds to the second-half of the international trade sequence of the economics PhD. Students are assumed to be comfortable with the first-year sequence in Micro, Macro and Econometrics. Taking the first-half of the international trade sequence of the economics PhD is recommended.

The class covers a range of topics, including production networks, global value chains, foreign direct investment, firm boundaries and incomplete contracts, trade policy, geoeconomics, imperfect competition, and industrial policy.

Grading and Assignments

The assignments of this class will be geared towards helping the student transitioning into doing her own research. Do not think of this as just something you have to do for a class, but rather as an opportunity for getting started, or pushing further, your own research.

Students will be graded according to the following mandatory assignments (grading weights in parenthesis):

1. Participation in Class (10%)

There are positive externalities from having a good discussion about research in class.

2. Quantitative Exercise (50%)

To prepare you for the own research phase, you can flexibly pick your project. It just needs to have some relationship to what we do in class. Some ideas:

2.1. Estimate tariff pass-through in response to the US-China trade war.

2.2. Ex-ante and Ex-Post analysis of trade models' predictions of the US-China tariffs. Do either a macro style (Caliendo and Parro) kind of analysis or pick a particular HS4 group of products and see if you can estimate export supply and import demand and make ex ante predictions and compare to ex post outcomes.
2.3. Follow the steps of Baqaee and Farhi (2020, QJE) to calculate the costs of mark-ups in the US (Note: this requires compustat data)
2.4. Do something descriptive with sectoral input-output data from the BEA (e.g., trends in the intermediate input use, volatility in the Domar weight, labor shares, ...)

First part (Define a question and have some summary statistics.) due January 24
Second part (intermediate report) due February 28
Third part (complete report) due March 28

3. In-class presentation (20%) (February 26)

In week 7, we will cover recent research on Geoeconomics. Please prepare a 30-40min presentation of one of these papers. We will assign papers by January 24 – please let me know your preferences beforehand. Depending on the number of students enrolled in class, this will be a single or joint presentation.

4. Referee Report (20%)

Write a referee report of a job market paper of Trade from the last job market. We will distribute a list of suggestions for this.

Your report should include: (a) summary of the main features of the paper; (b) description of its contribution to the literature; (c) its main strengths and weaknesses; (d) and your recommendations to improve it.

The report should have no more than 4 pages.

Deadline: April 15th.

Books (and online lectures)

Trade, macro, and IO, all have become quite demanding fields for the researcher. Many successful researchers are both good applied theorists and empiricists. It is also helpful to have a good understanding of computational methods.

For a second year PhD student interested in International Trade I recommend the following books:

Computational stuff:

- Judd (1998): Numerical Methods in Economics, MIT press
- Miranda and Faeckler (2002): Applied Computational Economics and Finance, MIT press [**highly recommended** for second year PhD students; it requires very little initial knowledge and takes you quite far]
- I **highly recommend** you watch this lecture by Michael Keane on practical issues in structural estimation at UChicago in November 2015. There are many very important suggestions contained in his lecture. Following these can save you months of your research time. They also help to ensure that you produce a good piece of research: https://www.youtube.com/watch?v=0hazaPBAYWE

Industrial Organization:

- Train (2009): Discrete Choice Methods with Simulation, Cambridge University press [**highly recommended**]
- Anderson, De Palma, and Thisse (1992): Discrete Choice Theory of Product Differentiation, MIT press

Statistics / Distributions:

- Kotz and Nadarajah (2000): Extreme Value Distributions: Theory and Applications, ICP
- Feller (1986): An Introduction to Probability Theory and Its Applications, Vol. I & II, Wiley

Econometrics:

- Cameron and Trivedi (2005): Microeconometrics: Methods and Applications, Cambridge University Press
- Hayashi (2000): Econometrics, Princeton University Press

Trade:

- Antras (2015): Global Production: Firms, Contracts, and Trade Structure
- Feenstra (2015): Advanced International Trade, Princeton University Press

- Helpman (2011): Understanding Global Trade
- See also the excellent lecture notes by Allen and Arkolakis: http://www.econ.yale.edu/~ka265/teaching/GradTrade/notes/ClassNotes.pdf
- You might also be interested in Pol Antras recent paper: "Unchartered Waters of International Trade" https://scholar.harvard.edu/antras/publications/uncharted-waters-international-trade

Topics

1. **Production Networks**

WEEK 1

Review articles of the literature:
Carvalho and Tahbaz-Salehi (2019): "Production Networks: A Primer," *Annual Review of Economics*
Bermard and Moxnes (2018): "Networks and Trade," *Annual Review of Economics*
Antras and Chor (2022): "Global Value Chains," Handbook of International Economics, Volume 5
Baqaee and Rubbo (2022): "Micro Propagation and Macro Aggregation," *Annual Review of Economics*

Closed economy macro:
Gabaix (2011): "The Granular Origins of Aggregate Fluctuations," *Econometrica*
Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi (2012): "The network origins of aggregate fluctuations," *Econometrica*
Hulten (1978): "Growth Accounting with Intermediate Inputs," *Review of Economic Studies*
Baqaee and Farhi (2019): "The Macroeconomic Impact of Microeconomic Shocks: Beyond Hulten's Theorem," *Econometrica*
Baqaee and Farhi (2020): "Productivity and Misallocation in General Equilibrium," *Quarterly Journal of Economics*
Baqaee and Farhi (2021): "Entry versus Rents," WP

Open economy macro:
Antras and De Gortari (2020): "On the Geography of Global Value Chains," *Econometrica*
Caliendo and Parro (2015): "Estimates of the Trade and Welfare Effects of NAFTA," *Review of Economic Studies*
Baqaee and Farhi (2024): "Networks, Barriers, and Trade," *Econometrica*

Giovanni and Levchenko (2012): "Country Size, International Trade, and Aggregate Fluctuations in Granular Economies," *Journal of Political Economy*
Giovanni, Levchenko, and Mejean (2018): "The Micro Origins of International Business-Cycle Comovement," *American Economic Review*
Gaubert and Ithskhoki (2021): "Granular Comparative Advantage," *Journal of Political Economy*

WEEK 2

Endogenous networks :
Eaton, Jinkins, Tybout, and Xu (2016): "Two-Sided Search in International Markets," WP
Oberfield (2018): "A Theory of Input-Output Architecture," *Econometrica*
Bernard, Moxnes, and Saito (2019): "Production Networks, Geography and Firm Performance," *Journal of Political Economy*
Lim (2018): "Endogenous Production Networks and the Business Cycle," WP
Dhyne, Kikkawa, Kong, Mogstad, and Tintelnot (2023): "Endogenous Production Networks with Fixed Costs," *JIE*
Huneeus (2018): "Production Network Dynamics and the Propagation of Shocks," WP
Bernard, Dhyne, Magerman, Manova, and Moxnes (2022): "The Origins of Firm Heterogeneity: A Production Network Approach," Journal of Political Economy
Demir, Fieler, Yang, and Xu (2024): O-Ring Production Networks, *JPE*
Miyauchi (2023): Matching and Agglomeration: Theory and Evidence from Japanese Firm-to-Firm Trade, WP
Taschereau-Dumouchel (2020): "Cascades and Fluctuations in an Economy with an Endogenous Production Network," WP
Baqaee (2018): "Cascading Failures in Production Networks," *Econometrica*

Facts and Empirics of shock propagation:
Acemoglu, Akcigit, and Kerr (2016): "Networks and the Macroeconomy: An Empirical Exploration," *NBER Macroeconomics Annual*
Barrot and Sauvagnat (2016): "Input Specificity and the Propagation of Idiosyncratic Shocks in Production," *Quarterly Journal of Economics*
Boehm, Flaaen, and Pandalai-Nayar (2019): "Input Linkages and the Transmission of Shocks: Firm-Level Evidence from the 2011 Tohoku Earthquake," *Review of Economics and Statistics*
Carvalho, Nirei, Saito, and Tahbaz-Salehi (2021): "Supply Chain Disruptions: Evidence from the Great East Japan Earthquake," *Quarterly Journal of Economics*
Dhyne, Kikkawa, Mogstad, and Tintelnot (2021): "Trade and Domestic Production Networks," *Review of Economic Studies*

Other:

Kikkawa, Magerman, and Dhyne (2020): "Imperfect Competition in Firm-to-Firm Trade," WP
Eaton, Kortum, and Kramarz (2018): "Firm-to-Firm Trade: Imports, Exports and the Labor Market," WP
Johnson and Noguera (2012): "Accounting for Intermediates: Production Sharing and Trade in Value Added," *Journal of International Economics*
Johnson (2018): "Measuring Global Value Chains," *Annual Review of Economics*
Pol Antràs, Davin Chor, Thibault Fally, and Russell Hillberry (2012): "Measuring the Upstreamness of Production and Trade Flows," AER P&P

2. Theories of the firm, Incomplete Contracts and Firms' Integration Strategies, Offshoring

WEEK 3

Main Readings:
Gibbons (2005): "Four formal(izable) theories of the firm?" *Journal of Economic Behavior & Organization*
Grossman and Helpman (2002): "Integration versus outsourcing in industry equilibrium," *Quarterly Journal of Economics*
Pol Antras (2003): "Firms, Contracts, and Trade Structure," *Quarterly Journal of Economics*
Antras, Fort, and Tintelnot (2017): "The Margins of Global Sourcing: Theory and Evidence from US Firms," *American Economic Review*

Additional Readings:
Antras and Helpman (2004): "Global Sourcing," *Journal of Political Economy*
Antras (2015): "Global Production: Firms, Contracts, and Trade Structure," *Princeton University Press*
Coase (1937): "The Nature of the Firms," *Economica*
Williamson (1985): "The Economic Institutions of Capitalism," Monograph
Grossman and Hart (1986): "The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration," *Journal of Political Economy*
Nunn (2007): "Relationship-Specificity, Incomplete Contracts and the Pattern of Trade," *Quarterly Journal of Economics*
Yi (2003): "Can Vertical Specialization Explain the Growth of World Trade?" *Journal of Political Economy*
Grossman and Rossi-Hansberg (2008): "Trading Tasks: A Simple Theory of Offshoring," *American Economic Review*
Rodriguez-Clare (2010): "Offshoring in a Ricardian World," *American Economic Journal: Macroeconomics*

Grossman and Rossi-Hansberg (2012): "Task Trade between Similar countries," *Econometrica*
Blaum, Peters, and Lelarge (2018): "The Gains from Input Trade in Firm-Based Models of Importing," *American Economic Journal: Macroeconomics*
Boehm and Oberfield (2020): "Misallocation in the Market for Inputs: Enforcement and the Organization of Production," QJE

3. Multinational Production and Multinational Firms:

WEEKS 4 & 5

Introduction:
Antras and Yeaple (2014): "Multinational Firms and the Structure of International Trade," Handbook of International Economics, Volume 4
Brainard (1997): "An Empirical Assessment of the Proximity-Concentration Trade-off Between Multinational Sales and Trade," *American Economic Review*
Helpman, Melitz, and Yeaple (2004): "Export Versus FDI with Heterogeneous Firms," *American Economic Review*
Helpman (1984): "A Simple Theory of International Trade with Multinational Corporations," *Journal of Political Economy*
Alfaro and Charlton (2009): "Intra-industry Foreign Direct Investment," *American Economic Review*
Ramondo, Ruhl, and Rappaport (2016): "Intrafirm Trade and Vertical Fragmentation in U.S. Multinational Corporations," *Journal of International Economics*
Ramondo, Rodriguez-Clare, and Tintelnot (2015): "Multinational Production: Data and Stylized Facts," *American Economic Review P&P*
Navaretti and Venables: Multinational Firms in the World Economy
Yeaple (2009): "Firm heterogeneity and the structure of U.S. multinational activity," *Journal of International Economics*
Nocke and Yeaple (2008): "An Assignment Theory of Foreign Direct Investment," *Review of Economic Studies*
Burstein and Monge (2009): "Foreign Know-How, Firm Control, and the Income of Developing Countries," *Quarterly Journal of Economics*
Irarrazabal, Moxnes, Opromolla (2013): "The Margins of Multinational Production and the Role of Intrafirm Trade," *Journal of Political Economy*
Keller and Yeaple (2013): "The Gravity of Knowledge," *American Economic Review*

Multi-country models of trade and multinational production:
Ramondo and Rodriguez-Clare (2013): "Trade, Multinational Production, and the Gains from Openness," *Journal of Political Economy*
Arkolakis, Ramondo, Rodriguez-Clare, and Yeaple (2018): "Innovation and Production in the Global Economy," *American Economic Review*

Tintelnot (2017): “Global Production with Export Platforms,” *Quarterly Journal of Economics*

Miscellaneous topics: Barriers to knowledge transfers, Spillovers, Frictions and Preferences:
Greenstone, Hornbeck, and Moretti (2010): “Identifying Agglomeration Spillovers: Evidence from Winners and Losers of Large Plant Openings,” *Journal of Political Economy*
Muendler and Becker (2010): “Margins of Multinational Labor Substitution,” *American Economic Review*
Giroud (2013): “Proximity and investment: Evidence from plant-level data,” *Quarterly Journal of Economics*
Aitken and Harrison (1999): “Do Domestic Firms Benefit from Direct Foreign Investment? Evidence from Venezuela,” *American Economic Review*
Javorcik (2004): “Does Foreign Direct Investment Increase the Productivity of Domestic Firms? In Search of Spillovers Through Backward Linkages,” *American Economic Review*
Setzler and Tintelnot (2021): The Effects of Foreign Multinationals on Workers and Firms in the United States, *Quarterly Journal of Economics*
Alfaro-Urena, Manelici, and Vasquez (2020): The Effects of Joining Multinational Supply Chains: New Evidence from Firm-to-Firm Linkages, WP
Alfaro-Urena, Manelici, and Vasquez (2019): The Effects of Multinationals on Workers: Evidence from Costa Rica, WP
Bilir and Morales (2020): “Innovation in the Global Firm,” *Journal of Political Economy*
Bloom, Schankermann, and Van Reenen (2013): “Identifying technology spillovers and product market rivalry,” *Econometrica*
Muendler and Becker (2010): “Margins of multinational labor substitution,” *American Economic Review*
Boehm, Flaaen, and Pandalai-Nayar (2020): ``Multinationals, Offshoring, and the Decline of U.S. Manufacturing, *Journal of International Economics*
Kovak, Oldenski, and Sly (2019): ``The Labor Market Effects of Offshoring by U.S. Multinational Firms,” *Review of Economics and Statistics*
Head and Mayer (2019): “Brands in Motion: How Frictions Shape Multinational Production,” *American Economic Review*
Cosar, Grieco, Li, Tintelnot (2018): “What Drives Home Market Advantage?” *Journal of International Economics*
Wang (2021): “Headquarters Gravity: How Multinationals Shape International Trade,” *Journal of International Economics*

4. Trade Policy & Geoeconomics

WEEKS 6 & 7

Main Readings:

Bagwell and Staiger (1999): "An Economic Theory of GATT," *Journal of Political Economy*

Costinot, Rodriguez-Clare, and Werning (2020): "Micro to Macro: Optimal Trade Policy with Firm Heterogeneity," *Econometrica*

Costinot and Werning (2019): "Lerner Symmetry: A Modern Treatment," *American Economic Review: Insights*

Costinot, Donaldson, Vogel, and Werning (2015): "Comparative Advantage and Optimal Trade Policy," *Quarterly Journal of Economics*

Ossa (2014): "Trade Wars and Trade Talks with Data," *American Economic Review*

Lashkaripour (2021): "The Cost of a Global Tariff War: A Sufficient Statistics Approach," *Journal of International Economics*

Handley and Limao (2017): "Policy Uncertainty, Trade, and Welfare: Theory and Evidence for China and the U.S." *American Economic Review*

McLaren (1997): "Size, Sunk Costs, and Judge Bowker's Objection to Free Trade," *American Economic Review*

Recent US Trade Conflicts:

Amiti, Redding, and Weinstein (2019): "The Impact of the 2018 Tariffs on Prices and Welfare," *Journal of Economic Perspectives*

Fajgelbaum, Goldberg, Kennedy, and Khandelwal (2020): "The Return to Protectionism," *Quarterly Journal of Economics*

Flaaen, Hortaçsu, and Tintelnot (2020): "The Production Relocation and Price Effects of US Trade Policy: The Case of Washing Machines," *American Economic Review*

Cavallo, Gopinath, Neiman, and Tang (2021): "Tariff Pass-Through at the Border and at the Store: Evidence from US Trade Policy," *American Economic Review: Insights*

Geoeconomics

Becko (2023), A Theory of Economic Sanctions as Terms-of-Trade Manipulation, Journal of International Economics

Becko and O'Connor (2024), "Strategic (Dis)Integration," Working Paper

Alekseev and Lin, "Trade Policy in the Shadow of Conflict: The Case of Dual-Use Goods," Working Paper

Kooi (2024), "Power and Resilience: An Economic Approach to National Security Policy," Working Paper

5. Search and Matching Frictions in Firm-to-Firm Linkages

WEEK 8

Eaton, Jinkins, Tybout, and Xu (2016): “Two-Sided Search in International Markets,” WP
Demir, Fieler, Xu and Yang (2024): “O-Ring Production Networks,” Journal of Political Economy
Arkolakis, Huneeus and Miyauchi (2023): “Production Network Formation, Trade and Welfare,” WP

6. Labor Markets in Production Networks

WEEK 9

Eaton, Kortum, and Kramarz (2018): “Firm-to-Firm Trade: Imports, Exports and the Labor Market,” WP
Huneeus, Kim, Kroft (2023): “Earnings Inequality in Production Networks,” WP
Dhyne, Kikkawa, Komatsu, Mogstad and Tintelnot (2022): “Foreign Demand Shocks to Production Networks: Firm Responses and Worker Impacts,” WP
Alfaro-Ureña, Manelici, Vasquez (2021): “The Effects of Multinationals on Workers: Evidence from Costa Rican Microdata,” WP

7. Imperfect Competition in Production Networks

WEEK 10

Kikkawa, Magerman, and Dhyne (2022): “Imperfect Competition in Firm-to-Firm Trade,” JEEA
Alviarez, Fioretti, Kikkawa and Morlacco (2022): “Two-Sided Market Power in Firm-to-Firm Trade,” WP
Burstein, Cravino, Rojas (2024): "Input Price Dispersion Across Buyers and Misallocation," WP

8. Industrial Policy in Production Networks

WEEK 11

Liu (2019): “Industrial Policies in Production Networks,” Quarterly Journal of Economics
Liu and Ma (2023): “Innovation Networks and R&D Allocation,” WP
Reka Juhasz's review papers

9. Heterogeneous Households in Production Networks

WEEK 12

Baqaee and Burstein (2-3 papers)
Atkin, Bernardac, Donaldson, Garg, Huneeus (2024): "The Incidence of Distortions," WP

10. Facts on Production Networks

WEEK 13

Pol Antràs, Davin Chor, Thibault Fally, and Russell Hillberry (2012): "Measuring the Upstreamness of Production and Trade Flows," AER P&P
Lim (2018): "Endogenous Production Networks and the Business Cycle," WP
Bernard, Moxnes, and Saito (2019): "Production Networks, Geography and Firm Performance," Journal of Political Economy
Huneeus (2018): "Production Network Dynamics and the Propagation of Shocks," WP
Khanna, Morales and Pandalai-Nayar (2022): "Supply Chain Resilience: Evidence from Indian Firms," WP
Acemoglu, Akcigit, and Kerr (2016): "Networks and the Macroeconomy: An Empirical Exploration," NBER Macroeconomics Annual
Barrot and Sauvagnat (2016): "Input Specificity and the Propagation of Idiosyncratic Shocks in Production," Quarterly Journal of Economics
Boehm, Flaaen, and Pandalai-Nayar (2019): "Input Linkages and the Transmission of Shocks: Firm-Level Evidence from the 2011 Tohoku Earthquake," Review of Economics and Statistics
Carvalho, Nirei, Saito, and Tahbaz-Salehi (2021): "Supply Chain Disruptions: Evidence from the Great East Japan Earthquake," Quarterly Journal of Economics
Dhyne, Kikkawa, Mogstad, and Tintelnot (2021): "Trade and Domestic Production Networks," Review of Economic Studies